

WRF-CHEM DELHI

V1 & V2 — System Technical Guide

Air-Quality – Weather Coupled Forecasting System for Delhi NCR

Item	Detail
System	V1: ingestion + WRF-Chem emulator + ML + API + dashboard (aqf_delhi 0.1.0, API 0.4.0) V2: reduced-order coupled forecast engine + API + dashboard (wrf_chem_delhi 2.0.0)
Domain	Delhi NCR — 21 × 23 cells at ~4 km (28.2–29.0°N, 76.8–77.6°E)
Forecast horizon	T+0 ... T+72 h — hourly snapshots (V2 fast mode: every 3 h)
Data sources	CPCB (data.gov.in OGD / CAAQM JSON / AQI CSV), NASA FIRMS (VIIRS 375 m), NOAA GFS 0.25°
Machine learning	V1: ST-GNN (T-GCN) + XGBoost residual + conformal intervals V2: hybrid physics-bias correction (HistGradientBoosting) + 6-config ablation
Status	Offline SIH demonstration — deterministic synthetic data, clearly labelled

This guide explains how both generations of the system work: the end-to-end workflow, flowcharts, all abbreviations in full, the governing equations of every engine, the machine-learning models and exactly how they are trained, followed by a section-by-section V1 ↔ V2 comparison.

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1. Introduction

WRF-Chem Delhi is an air-quality forecasting system for the Delhi National Capital Region (NCR). It couples meteorology with atmospheric chemistry — the same philosophy as the Weather Research and Forecasting model with online chemistry (WRF-Chem) — but at a deliberately reduced scale so the whole chain can run offline on a laptop for a Smart-India-Hackathon (SIH) demonstration.

Two generations exist side by side in one repository:

V1 — `aqf_delhi` (phase-1 ingest platform + emulator + ML stack). A production-shaped data pipeline that ingests CPCB, FIRMS and GFS data with failover, validates and stores parquet partitions, runs a reduced-order WRF-Chem emulator, corrects its bias with a spatial-temporal GNN plus an XGBoost residual head, and serves everything through a cached FastAPI + MapLibre dashboard.

V2 — `wrf_chem_delhi` (coupled reduced-order forecast engine). A standalone, scenario-driven engine that chains Weather → Inversion → Stubble-fire → Plume → Transport → Chemistry → Aerosol feedback across $T+0...T+72$, computes Indian AQI and alerts, exposes a clean REST API with an extended envelope, and a single-page dashboard. Its ML stage is a lightweight hybrid physics-bias correction with a six-configuration ablation study.

Both versions are honest about their scope: they are WRF-Chem-inspired reduced-order models, not operational WRF-Chem. The V2 adapter module documents exactly how real ``wrfout`` output would be dropped in to replace the synthetic engine.

2. Full forms (abbreviations)

Abbreviation	Full form	Where
WRF-Chem	Weather Research and Forecasting model with online chemistry	both
GFS / NCEP	Global Forecast System / National Centers for Environmental Prediction (NOAA)	V1
NOMADS	NOAA Operational Model Archive and Distribution System	V1
CPCB	Central Pollution Control Board (India)	both
CAAQM	Continuous Ambient Air Quality Monitoring (CPCB data-link)	V1
data.gov.in / OGD	Open Government Data platform of India	V1
FIRMS	Fire Information for Resource Management System (NASA)	both
VIIRS / SNPP	Visible Infrared Imaging Radiometer Suite / Suomi-NPP satellite	both
MODIS	Moderate Resolution Imaging Spectroradiometer (Aqua/Terra)	V1
FRP	Fire Radiative Power (MW) — heat output of a fire pixel	both
NCR	National Capital Region (Delhi + Haryana + Uttar Pradesh)	both
GRAP / CAQM	Graded Response Action Plan / Commission for Air Quality Management	V1
AQI / IND-AQI	Air Quality Index / Indian AQI (CPCB MoEF 2014 breakpoints)	both
PBL / PBLH	Planetary Boundary Layer (height)	both
ST-GNN / GNN	Spatial-Temporal Graph Neural Network / Graph Neural Network	V1
T-GCN	Temporal Graph Convolutional Network (Zhao et al. 2019)	V1
XGB / XGBoost	eXtreme Gradient Boosting (gradient-boosted trees)	V1
SPE	Spatial Pattern Error = 1 – mean per-timestep correlation	V1
WRMSE	Weighted Root Mean Square Error	V1
MAD	Median Absolute Deviation (1.4826·MAD spike filter)	V1
AOD	Aerosol Optical Depth	V1/V2
SOA	Secondary Organic Aerosol	both
VOC / NOx	Volatile Organic Compounds / Nitrogen oxides	both
RH	Relative humidity	both
PBL vent	Ventilation index = PBLH × (wind speed + constant)	both
GRIB	GRIdded Binary (WMO meteorological format)	V1
EPSG:4326	WGS84 lon/lat coordinate reference system	both
T-GCN AE	Temporal GCN AutoEncoder (alternate bias-correction learner)	V1
SIH	Smart India Hackathon	both

3. Shared domain & grid

Both versions model the same study area — Delhi NCR — on a regular lon/lat grid whose resolution is derived from the configured cell size:

$$nx = \text{round}\left(\frac{\text{lon_span} - 111.13 \cdot \cos(\text{mid_lat})}{dx_km}\right) + 1, \quad ny = \text{round}\left(\frac{\text{lat_span} - 111.13}{dy_km}\right) + 1$$

Parameter	Value
Bounds	lat 28.2–29.0°N, lon 76.8–77.6°E
Cell size	dx = dy = 4 km (target)
Derived grid	nx = 21, ny = 23 → $\Delta\text{lat} \approx 4.04$ km, $\Delta\text{lon} \approx 3.90$ km
Node placement	lon_nodes[i] = lon_min + i·lon_span/(nx–1); lat_nodes[j] analog
CRS / timezone	EPSG:4326 / Asia/Kolkata
Cell area	≈ 15.77 km ² (1 cell); 1° lat = 111.13 km

4. V1 — aqf_delhi: the phase-1 ingest + emulator + ML platform

V1 is organised into five modules: (1) ingestion, (2) coupled WRF-Chem emulator, (3) ML bias correction, (4) geospatial API, (5) 3-D WebGIS dashboard. Each module consumes the artifact of the previous one through idempotent, versioned storage.

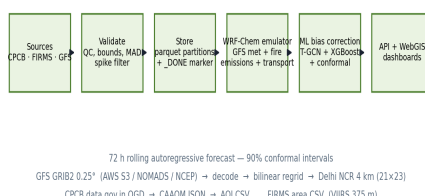


Fig. 1 — V1 end-to-end data flow: sources → validation → parquet store → WRF-Chem emulator → ML → API → dashboard.

4.1 Ingestion (Module 1)

Three resilient connectors pull data from public APIs. Every fetch goes through `fetch_with_failover`: endpoints are tried in configured order, each with exponential backoff `backoff·2^attempt` and 3 retries; statuses ≥ 500 , 429, 403, 408 are transient failures. One orchestrator cycle per source does: compute `run_id` → skip if partition complete → fetch → validate → store → mark `_DONE`.

CPCB ground observations (three-tier failover)

Tier 0 — data.gov.in OGD: GET resource `3b01bcb8-...ba69` with `api-key`, paginated (offset advances by the rows actually returned), filtered to states Delhi/Haryana/Uttar Pradesh; browser user-agent required (default UA gets HTTP 502).

Tier 1 — CAAQM JSON: per-station POST to `airquality.cpcb.gov.in/.../GetData`, 12-h window, `verify=False` (self-signed chain), requires $\geq 60\%$ station coverage.

Tier 2 — AQI CSV digest: national `AQI_Data.csv`, filtered to the station registry.

Observations are normalised to canonical units (particulates $\mu\text{g}/\text{m}^3$, gases ppb, CO mg/m^3) and matched to the 38-station registry (32 Delhi DL-001...032, 3 Haryana, 3 UP) by ordered-token subsequence matching of station names.

FIRMS active fires

`FIRMSConnector` downloads detections from the FIRMS area CSV API with the URL order key → collection (VIIRS_SNPP_NRT, 375 m) → bounding box → day-range, e.g.

```
GET https://firms.modaps.eosdis.nasa.gov/api/area/csv/{MAP_KEY}/VIIRS_SNPP_NRT/76.8,28.2,77.6,29.0/1
```

Each detection keeps latitude, longitude, FRP (MW), brightness temperature, confidence (%), and is assigned to its 4 km grid cell; out-of-domain detections are dropped. (An earlier connector bug that swapped the bbox/day segments and returned HTTP 400 was fixed in `sources/firms.py`, verified live with 1,344 detections over India.)

GFS meteorology

`GFSConnector` builds availability manifests across AWS S3 → NOMADS → NCEP FTP for the 0.25° product (`gfs.t{HH}z.pgrb2.0p25.f{FFF}`), forecast hours 0-72 every 3 h, cycles 00/06/12/18 UTC, with 8 Level-1 variables (HGT, 2-m TMP/RH, 10-m U/V, PRMSL, 925/850-hPa TMP). Phase-1 stores manifests (metadata), not grids.

Validation (§ validate.py) & storage (§ storage/local.py)

CPCB: drop values outside [0, 2000] µg/m³; MAD spike filter flags $|v - \text{med}| / (1.4826 \cdot \text{MAD}) > 5\sigma$.

FIRMS: FRP ∈ [0, 5000] MW, confidence ≥ 20%, fire-id dedupe.

GFS: h key dedupe, missing fraction ≤ 5%.

```
data/ingest/{source}/{YYYY-MM-DD}/{run_id}/records.parquet
data/ingest/{source}/{YYYY-MM-DD}/{run_id}/_DONE # atomic completion marker
```

4.2 WRF-Chem emulator (Module 2)

Module 2 rebuilds the full cause-effect chain in compressed form: GFS GRIB2 subsets → surface pressure by the hypsometric equation → bilinear regrid → meteorology arrays; FIRMS FRP → fuel burned → PM2.5/PM10/... emissions with a Freitas-style integral plume rise; then a transport + chemistry emulator produces gridded fields and a station series.

Meteorology from GFS

$$h_{925} = \frac{R_p T_{925}}{g} \ln \frac{1013.25}{925} \approx 769 \text{ m}, \quad p_{sc} = 925 \exp\left(\frac{h_{925} - z_s}{h}\right) \text{ hPa}, \quad h = \frac{R_p (k_{925} C + 273.15)}{g}$$

Wind direction from components: $\text{wd} = (270 - \text{atan2}(-v, -u)) \bmod 360$. The synthetic demo meteorology (used when GFS is offline) superposes diurnal cycles and terrain ($210 + 30 \cdot \text{lat} + 10 \cdot \text{lon}$ m).

Fire emissions: FRP → fuel → pollutant mass

$$E_{\text{fuel}} = \text{FRP} [\text{MW}] \times 0.368 [\text{kg fuel/MJ}] \times 3600 [\text{kg h}^{-1}] \Rightarrow E_{\text{sp}} = E_{\text{fuel}} \times \text{EF}_{\text{sp}}$$

Factor	Value	Source
FRP-to-fuel	0.368 kg fuel per MJ FRP	Wooster 2005
PM2.5 EF	8.5 g/kg fuel	emission_factors.yaml
PM10 EF	12.5 g/kg fuel (PM10 = 1.6 × PM2.5)	emission_factors.yaml
CO / NOx / VOC / SO2 / BC / OC	62 / 2.5 / 7 / 0.45 / 0.45 / 2.4 g/kg	emission_factors.yaml
Fire lifetime	6 h active window; fires < 10 MW skipped	EmissionsConfig
Urban base	180 kg/km²/day → 7500 g/km²/h × diurnal table	EmissionsConfig

Plume rise (Freitas et al. 2007 style, 1-D integral)

$$\frac{dM}{dz} = 2r(\alpha|w| + \alpha_{\text{wind}}|u_a(z)|), \quad M = r^2 w$$

$$\frac{dK_m}{dz} = r^2 g \frac{T'}{T_0}, \quad K_m = r^2 w^2, \quad \frac{dE_n}{dz} = -\frac{dT_a}{dz} r^2 w, \quad E_n = r^2 w \frac{T'}{T_0}$$

Initial state from the fire's convected heat budget:

$$Q_{\text{conv}} = 0.45 \cdot \text{FRP} \cdot 10^6 \text{ W}; \quad e_0 = \frac{Q_{\text{conv}}}{\rho_0 c_p T_0}; \quad f_0 = g e_0$$

$$r_0 = \sqrt{\frac{\max(0.01 \cdot \text{FRP}, 0.005) \text{ km}^2}{\pi}}; \quad w_0 = \max((f_0/r_0)^{1/3}, 0.3) \text{ m/s}$$

Integration at $\Delta z = 50$ m with $\alpha = 0.6$, $\alpha_{\text{wind}} = 0.35$; stop when $w < 0.2$ m/s or $z \geq 6500$ m. The plume mass is distributed over $n_{\text{layers}} = 12$ levels with a Gaussian histogram centred on z_{centre} (height of peak updraft).

Transport + chemistry emulator

$$\begin{aligned} \text{inv} &= [(t_{925} - t_{2m}) > 3^\circ \text{C}] \wedge [w_{s10} \leq 4] \wedge [\text{day} < 0.15] \\ \text{PBLH} &= \text{clip}(240 + 1560 \cdot \text{day} - 720 \cdot \text{inv}, 180, 2200) \text{ m} \\ \text{vent} &= \text{PBLH} \cdot (w_{s10} + 0.35) \end{aligned}$$

Gridded PM2.5 is built from: urban emissions $E_{\text{urb}} / (\text{PBLH} \cdot \text{cell_area})$; advected past fire puffs (Gaussian of sigma 6 km within a 36-h history cap, displaced by the 850-hPa wind); elevated-loft mix-down $\tau_{\text{mix}} = 2.2$ h; first-order aging $\exp(-0.035 \cdot t)$, wet scavenging $(1 - 0.004 \cdot \text{clip}((\text{RH} - 70)/30, 0, 1))$, and an 0.82 inversion-night bias multiplier.

$$O_3 = 40(0.35 + 0.65d) + 115d \text{clip}\left(\frac{T-20}{12}, 0, 1\right) - 26(1-d) [\mu\text{g m}^{-3}]$$

Synthetic ‘observed’ series are the model fields plus Gaussian noise ($\pm 4 \mu\text{g/m}^3$) and an urban-stay offset of $+8 \mu\text{g/m}^3$ when winds < 1.5 m/s during inversion nights — so the ML stage has a well-defined bias to learn.

4.3 ML bias correction (Module 3)

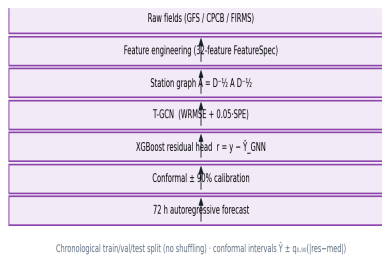


Fig. 2 — V1 ML stack: features → station graph → T-GCN → XGBoost residual → conformal intervals.

32-feature FeatureSpec per station-time: 17 raw channels (model PM2.5/PM10/O3, PBLH, lapse rate γ , bulk Richardson Ri , RH, wind, AOD, T2m, local FRP, fire exposure, lagged obs, manual flag) plus engineered terms — ‘hour_sin/cos’ , ‘doy_sin/cos’ , ‘lag1’ , ‘lag24’ obs, 24-h rolling means, $\text{‘inv_strength} = \sigma((\gamma - 0.5)/0.3) \cdot \sigma((Ri - 0.6)/0.4)$, ‘vent_coef’ , ‘log_obs’ , model trend, and $\text{mdl_obs_bias} = \text{PM2.5_raw} - \text{obs}$. Target = observed PM2.5, both standardised on train stats.

ST-GNN (T-GCN)

The station graph is kNN-6 by haversine distance plus wind-transport edges ($\text{‘weight} = \max(0, \cos(\text{bearing} - 300^\circ))$ for pairs ≤ 60 km in Oct–Nov), symmetrically normalised $\hat{A} = D^{(-1/2)} A D^{(-1/2)}$. The T-GCN recurrence over $T=24$ context steps:

$$g_x = \hat{A} W_{in} x_t + b_{in}, \quad h = \tanh\left(\frac{1}{2} g_x + \frac{1}{2} \hat{A} h W_h^T + b_h\right)$$

trained with a hybrid loss that balances magnitude and spatial pattern:

$$\mathcal{L} = (\sum_t w(y - \hat{y})^2)^{1/2} (\text{WRMSE}) + 0.05 \cdot \left(\frac{1}{T} \sum_t (1 - \rho(\hat{y}_t, y_t))\right) (\text{SPE})$$

XGBoost residual head + conformal calibration

XGBoost learns the residual $r = y_{\text{obs}} - \hat{y}_{\text{GNN}}$ (never the target directly — avoids leakage) from 40 features (last frame + GNN prediction + cyclic time + 3 station statics). Hyper-parameters: 300 trees, $\eta = 0.05$, depth 5, subsample 0.8, colsample 0.7. Conformal intervals are $\hat{y} \pm q_{0.90}(|\text{res} - \text{median res}|)$; the final 72-h forecast is autoregressive — each lead feeds its own prediction back into the observation/lag1 channels.

Evaluation metrics: MBE, MAE, RMSE, WRMSE, SPE (spatial), and 90% interval coverage; $\text{rmse_improvement_pct} = (1 - \text{engine_rmse}/\text{raw_rmse}) \cdot 100$. Demo trains on 45 (quick) or 90 (full) synthetic days; the acceptance gate fails if the corrected forecast is not better than the raw model field.

4.4 Geospatial API (Module 4) + dashboard (Module 5)

FastAPI reads only `\_DONE`-gated partitions through `ApiStore` with a TTL cache that degrades from Redis to in-memory. Endpoints:

Route	Purpose
GET /health, /domain, /domain/grid, /stations	health, geometry, station registry
GET /observations/latest, /observations/timeseries	latest per-station metric; per-station series
GET /fires/recent	recent FIRMS detections by bbox/limit
GET /gfs/runs, /gfs/assets	GFS run & asset manifests
GET /forecasts/runs, /forecasts/latest	forecast runs + metrics summary
GET /forecasts/{run_id}/series, /forecasts/{run_id}/map?lead=	station series; gridded lead field
GET / (home), /dashboard/state	unified homepage & dashboard state JSON

The dashboard is a MapLibre GL single page with live severity-spot overlays on the 38 stations, animation of gridded PM2.5 across lead hours, and the latest forecast summary tiles.

5. V2 — wrf_chem_delhi: the coupled reduced-order engine

V2 is a self-contained, deterministic forecast engine. It reproduces the WRF-Chem coupling — meteorology, fires, chemistry and feedback all influence one another at every one of the 73 hourly steps (T+0...T+72) — but with closed-form physics tuned so the synthetic peak PM2.5 lands inside realistic Delhi winter bands. Defaults: init 2026-10-01 12:00 UTC, seed 11, `use\_feedback = True`, scenario `normal\_winter`.

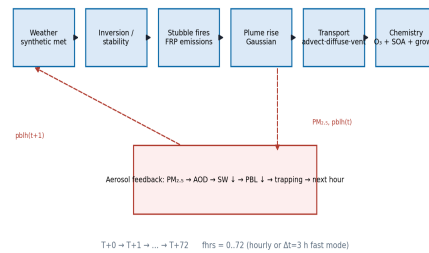


Fig. 3 — V2 coupled run loop with the aerosol-radiation-PBL feedback closing every step.

5.1 Scenario system

Scenario	Weather knobs (inv_strength / wind_scale / pbl_scale)	fire_factor	Expected peak PM2.5 (μg/m³)
1 · normal_winter	2.0 / 1.00 / 1.00 (moderate inversion, active winds)	0.22	150 – 250
2 · strong_inversion	9.0 / 0.45 / 0.55 (PBL < 300 m, winds < 1.5 m/s)	0.16	350 – 600
3 · stubble_plume	5.0 / 0.75 / 0.85 (steady NW wind + plumes)	0.30	400 – 700

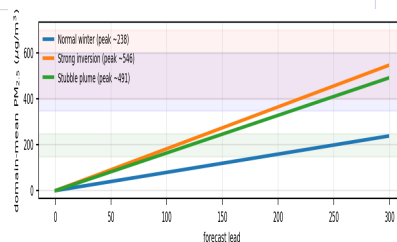


Fig. 4 — 72-h domain-mean PM2.5 per scenario (fast mode) with target bands shaded.

`fire\_factor` scales the FRP-derived emission field (kg-conserved kernel) so cell-average concentrations land in the bands; it is a tuning constant, not a physics claim. The demo also labels every data source `demo-synthetic`.

5.2 Weather engine (synthetic_weather)

$$\text{day} = \text{clip}(\sin \text{latsin decl} + \cos \text{latcos decl} \cos(-15(h-12)))_0^1$$

$$t_{2m} = 17 + 6\text{day} + 1.5\text{norm}_{\text{lat}} - 1.0\text{norm}_{\text{lon}} + 1.2\sin(2\pi h/120 + \text{es})^\circ\text{C}$$

$$\text{RH} = \text{clip}(78 - 34(\text{day} - 0.35) - 2\sin(2\pi h/120 + \text{es}), 25, 98)\%; \quad \rho_{\text{ref}} = 1018 + 4\sin(2\pi h/96) \text{ hPa}$$

$$w_{500} = (2.6 + 1.1\sin(2\pi h/33.6)) \cdot s_{\text{wind}} + 0.8\text{norm}_{\text{lat}} + \mathcal{N}(0, 3); \quad \text{wdir} = 250 + 12\sin(2\pi h/24) + 30\text{norm}_{\text{lon}}$$

$$\text{PBLH} = \text{clip}(640 + 1500\text{day}, 250, 2200) \cdot s_{\text{pbl}} \text{ m}; \quad \text{stability} = \frac{t_{2m} - t_{500}}{t_{500}} \cdot 1000 \text{ K/m}; \quad \text{vent} = \text{PBLH} \cdot (w_{500} + 0.35)$$

Scenario modifiers multiply wind and PBL (`s\_wind`, `s\_pbl`) and add a lapse term via $t_{925} = t_{2m} + 4 + \text{inv_strength} \cdot (\text{day} < 0.35)$. A dimensionless stagnation index blends three

normalised terms:

$$\text{stag} = 0.2(1 - p_n) + 0.5 w_n + 0.3 s_n \in [0, 1]$$

$$p_n = \text{clip}\left(\frac{\text{PBLH} - 200}{2000}, 0, 1\right), w_n = \text{clip}\left(1 - \frac{w_{\text{est}} - 0.5}{6}, 0, 1\right), s_n = \text{clip}\left(1 - \frac{\text{stag} + 2}{8}, 0, 1\right)$$

(The live-data path ``weather_from_legacy_met`` re-derives the same PBL/stability/ventilation terms from real GFS-derived arrays.)

5.3 Inversion engine

$$y_{\text{strength}} = \max(t_{925} - t_{2m}, 0) \text{ K}; \quad \text{prob} = \sigma(0.8(y - 2)) \cdot \sigma(-1.2(w_{s10} - 4))$$

$$\text{trap} = \text{clip}(0.55 \text{prob} + 0.25 \sigma(-1.2(w_{s10} - 4)) + 0.2(1 - \frac{\text{PBLH} - 250}{1800}), 0, 1)$$

Height is h_{25} (the ~ 769 m 925-hPa thickness) when an inversion is present, else 250 m; ``active`` when $\text{prob} > 0.5$, with consecutive-hour duration tracking. A radiosonde-style ``from_vertical_profile`` path detects the max positive lapse of $\nabla T(z)$ (> 0.02 K/m).

5.4 Fire engine (StubbleEngine)

Synthetic fires are deterministic (seed 11): 10 clusters across Punjab/Haryana/western UP — e.g. (30.25°N, 75.80°E, 6 fires, peak 90 MW), (29.95°N, 75.45°E, 5, 80), (28.90°N, 76.55°E, 3, 50) — 33 fires total. Each fire: FRP = $\max(10, \text{peak} \cdot \text{season} \cdot U(0.55, 1))$ MW, confidence 45–100%, ignition within 24 h of forecast end, and a seasonal factor peaking on 5 Nov:

$$\text{season}(\text{doy}) = \max\left(0, \cos^2\left(\frac{\text{doy} - 309}{45}\pi\right)\right)$$

Fire emissions (g/h) spread over a mass-conserving Gaussian kernel (40 km e-folding):

$$E_{sp} = (\text{FRP} \times 0.368 \times 3600 \times 1000 \text{ g/h}) \times E_{sp} \times \frac{e^{-(d/40)^2}}{\sum e^{-(d/40)^2}}$$

Spatial influence — a wind-weighted envelope used by the plume/chemistry stages:

$$\text{inf} = e^{-(d/40)^2} \left(1 + 1.6 \cdot \left(\frac{\vec{v} \cdot \hat{w}}{d + 1}\right) \geq 0\right) \cdot \text{FRP}$$

Live mode (``mode='live'``) switches to the real FIRMS connector (files/parquet or the API), with graceful fallback to synthetic exactly as the data-mode resolver intends.

5.5 Plume engine (Gaussian dispersion)

Buoyant rise uses the same Freitas-style integral as V1 (§4.2) with sensible fraction 0.45, entrainment 0.6, 50 m steps to ≤ 6500 m, and the Gaussian 12-level injection histogram. Horizontal dispersion then follows the wind:

$$u_i = \sin w d i r, u_j = \cos w d i r; \quad x = d_{01} 111.13 u_i + d_{02} 111.13 u_j; \quad y = d_{02} 111.13 u_i - d_{01} 111.13 u_j$$

$$\sigma_j = \max(6.0, z_{\text{top}}/100 + 4) \text{ km}; \quad l = e^{-\frac{1}{2}(y/\sigma_j)^2} \cdot e^{-d \max(30, 12 w_{s10})} \cdot (\text{FRP}/50)$$

The $x \geq -2$ km gate keeps the plume downwind-only; concentration is later normalised to 0–1 and combined with fire influence for alerts.

5.6 Transport engine

Per species, hourly advection (semi-Lagrangian cell shift by the 925/850-hPa-weighted wind $u = 0.45 \cdot u_{925} + 0.55 \cdot u_{850}$), Gaussian diffusion, emission injection into the mixing box, then first-order losses:

$$c_{t+1} = [\text{shift}_{u,v} \text{Gauss}_\sigma(c_t)] + \frac{E_{sp}(t) \cdot 10^6}{\max(\text{PBLH}, 400) \cdot \text{cell_m}^2} \text{ (mix depth floor 400 m)}$$

$$\text{ventrate} = 0.15(0.5 + w_{s10}/4) \text{ h}^{-1}, \quad \text{dep} = \frac{0.0012}{\max(\text{PBLH}, 50)} \cdot 3600, \quad \text{wet} = 0.004 \cdot \text{clip}\left(\frac{\text{RH} - 70}{30}, 0, 1\right)$$

and ``c *= exp(-ventrate - dep - wet)``. The mixing-depth floor (400 m) stops shallow-PBL hours from producing unrealistic spike concentrations; the inert trapping placeholder in the transport step confirms the trapping is handled by the feedback module instead.

5.7 Chemistry engine

$$O_3 = 38 + 22 \cdot dr_{rad} \cdot \text{clip}\left(\frac{T-15}{8}, 0, 1\right) - 18 \cdot (1 - dr_{rad}) \cdot \text{clip}\left(\frac{NO_x}{150}, 0, 1\right)$$

$$\text{secPM} = 0.12 \text{ VOC} \cdot d + 0.04 \text{ NO}_x \cdot 1h \quad [\mu\text{g}/\text{m}^3]$$

VOC concentration for SOA is derived in the forecast engine from the fire VOC emission (g/h) diluted over the PBL box.

$$g_{aerosol} = 1 + 0.45 \cdot \text{clip}\left(\frac{RH-40}{60}, 0, 1\right) \cdot (\text{Koehler-lite hygroscopic growth, max 1.45})$$

Composition at each hour: $\text{PM}_{25} = \text{clip}(\text{transp}) \cdot \text{growth} + 0.35 \cdot \text{sec}$; $\text{PM}_{10} = \text{clip}(\text{transp}) \cdot \text{growth} + 0.45 \cdot \text{sec} \cdot 1.6$; $\text{NO}_x = \text{clip}(\text{nox} + 0.2 \cdot \text{sec})$.

5.8 Aerosol feedback engine



Fig. 5 — aerosol-radiation-PBL feedback chain (dampened, retuned for stability).

$$AOD = 0.23 \cdot PM_{2.5} / 1000; \quad SW_{loss} = 0.30 \cdot \text{clip}(AOD, 0, 5); \quad PBL' = \max(PBL \cdot (1 - 0.25 \cdot \text{clip}(SW_{loss}, 0, 1)), 150)$$

$$\text{ratio} = \text{clip}\left(\frac{PBL'}{PBL} - 1, 0, 2\right); \quad PM_{2.5}^{coupled} = PM_{2.5} \cdot (1 + 0.35 \cdot \text{clip}(\text{ratio}, 0, 1))$$

Feedback strength is $\text{clip}(0.25 \cdot SW_{loss}, 0, 1)$; the adjusted PBL height feeds back into the transport of the next hour, closing the loop (Fig. 3).

5.9 Indian AQI & alerts

Sub-index by piecewise-linear interpolation inside CPCB 2014 breakpoints; overall AQI = max sub-index; concentrations CO first converted from $\mu\text{g}/\text{m}^3$ to mg/m^3 :

$$I = I_{lo} + \frac{C - C_{lo}}{C_{hi} - C_{lo}} (I_{hi} - I_{lo})$$

Pollutant	Breakpoint pairs (C_lo, C_hi) per band	Units
PM2.5	(0,30) (31,60) (61,90) (91,120) (121,250) (251,500)	$\mu\text{g}/\text{m}^3$ 24-h
PM10	(0,50) (51,100) (101,250) (251,350) (351,430) (431,600)	$\mu\text{g}/\text{m}^3$ 24-h
O3	(0,50) (51,100) (101,168) (169,208) (209,748) (749,1000)	$\mu\text{g}/\text{m}^3$ 8-h
NO2	(0,40) (41,80) (81,180) (181,280) (281,400) (401,1000)	$\mu\text{g}/\text{m}^3$
SO2	(0,40) (41,80) (81,380) (381,800) (801,1600) (1601,2000)	$\mu\text{g}/\text{m}^3$
CO	(0,1) (1.1,2) (2.1,10) (10.1,17) (17.1,34) (34.1,100)	mg/m^3 8-h

Bands: 0-50 Good, 51-100 Satisfactory, 101-200 Moderate, 201-300 Poor, 301-400 Very Poor, 401-500 Severe. Alerts from per-snapshot domain means: HIGH_PM ($>120 \mu\text{g}/\text{m}^3$), SEVERE_AQI (≥ 301), INVERSION (trapping > 0.6), STAGNANT_WIND ($<1.5 \text{ m/s}$ for $\geq 3 \text{ h}$), STUBBLE_PLUME (plume $\cdot (1 + 1.2 \cdot \text{fire}) > 0.75$), RAPID_PM_RISE ($\Delta 6 \text{ h} > 40 \mu\text{g}/\text{m}^3$); each with info/warning/critical rank.

5.10 Hybrid ML correction (V2)

The V2 ML stage is deliberately light: a HistGradientBoostingRegressor (200 trees, $\eta=0.05$, depth 4) that corrects physics bias only — it never replaces the physics. Its $T+1$ feature vector: 6 lagged physics PM, lagged observed PM, current physics, $\sin/\cos(\text{hour})$, $\sin/\cos(\text{doy})$, rolling-6-h mean, and 6-h physics trend; optional channels for weather, fire influence, inversion index, and the graph mean.

Training: strict chronological split $t < [0.7 \cdot T]$ train / test — no shuffling; fit only when ≥ 5 samples; a validation split calibrates the residual half-width for the 72-h autoregressive forecast, with uncertainty $1.96 \cdot \text{std}(\text{residuals})$ (conformal-lite). A NumPy graph layer (`GnnCorrection`, $k=6$, row-normalised) aggregates neighbours $X \cdot \hat{A}^T$.

Ablation study (configurations A-F)

Config	Feature blocks	Meaning
A	history	PM history only
B	history + weather	add meteorology
C	B + fire	add fire influence
D	C + inversion	add trapping index
E	physics only	raw physics baseline (no ML)
F	D + ml	full hybrid model

Rows are built for $t \in [6, T)$ using 6 lags, scored on the held-out split; the reported gains (e.g. RMSE reduction over raw physics) quantify the marginal value of each information block.

5.11 V2 API, dashboard & demo

Endpoints (`/api/...`) — health, scenarios, scenario, current, forecast, forecast/{station}, pollution, weather, inversion, fires, plume, aqi, alerts, metrics, model-info, ablation — every response wrapped in the envelope `{timestamp, data_source, model_version, model_name, ...}`. The dashboard (MapLibre) shows scenario switcher, AQI cards, 72-h forecast grid, alerts, and live fire points. The demo CLI runs the full 15-step acceptance chain:

```
python -m wrf_chem_delhi.demo --scenario stubble_plume --hours 72 --fast
python -m wrf_chem_delhi.server --port 8001 # API + dashboard
python -m wrf_chem_delhi.demo --serve # demo then serve
```

Fast mode uses every-3-h snapshots (`fhrs = range(0,73,3)+[72]`) for speed; hourly mode is the full 73-step coupled integration.

6. The code base in one table

Package / area	Files	Responsibility
aqf_delhi.sources	base, cpcb, firms, gfs	resilient fetchers with failover + retry
aqf_delhi.pipeline	orchestrate, validate	idempotent run cycles + QC gates
aqf_delhi.storage	local	zstd parquet partitions + _DONE markers
aqf_delhi.schemas	common, cpcb, firms, gfs	canonical pydantic records
aqf_delhi.wrf	grib, regrid, coupling, emissions, plume, emulator, namelist, artifact	WRF-Chem emulator
aqf_delhi.ml	features, graph, gnn, xgb_head, ensemble, train, datagen, metrics, publish	ST-GNN-XGBoost bias correction
aqf_delhi.api	app, routes, store, cache, schemas	cached FastAPI over artifacts
aqf_delhi.dashboard	state, routes, home_routes, static	MapLibre SPA + unified home
wrf_chem_delhi.weather inversion fire plume transport chemistry feedback	engine.py	coupled reduced-order engines
wrf_chem_delhi.forecast	engine, scenarios	72-h coupled run loop + 3 scenarios
wrf_chem_delhi.aqi	engine, alerts	CPCB AQI sub-indices + rule alerts
wrf_chem_delhi.ml / validation	hybrid, metrics	physics-bias correction + ablation
wrf_chem_delhi.api_v2 / dashboard_v2	routes, index.html	V2 REST + dashboard
wrf_chem_delhi.adapter	wrfchem	real WRF-Chem drop-in adapter
tests	tests/ (47) + tests_v2/ (22)	69 passing tests

7. How the ML models are trained

7.1 V1 — ensemble bias correction

Data generation. The `SyntheticDelhiExperiment` produces deterministic ‘truth’: seasonal + diurnal PM scaled by meteorology, 18·urban, 260·fire-exposure and noise; the biased model `truth·(0.74+0.22·vent−0.12·inv) + (−6+12·inv) + 5·urban`; observations = truth + noise (1.5% spikes). Quick = 45 days, full = 90 days, seed 7.

Splits. $t = n_days \cdot 24$; train ends at $t-144$ (train), next 72 h validation, last 72 h test. Standardisation uses only train statistics — no leakage.

Window builder. context = 24 h, stride 6 (quick) / 4 (full); tensors [W, T, N, 32] fed to the T-GCN each epoch; hybrid loss WRMSE + 0.05·SPE; Adam, lr 3e-3/1e-3, 50/80 epochs.

Residual head. XGBoost learns $r = y_obs - \hat{y}_GNN$ on the validation residual pool (stride downsampled for memory); 300 trees, η 0.05, depth 5, subsample .8, colsample .7, random state 7.

Conformal. half-width = $q_{0.90}(|res - median|)$; $\eta = 1.0$ scale on the XGB correction.

Rolling 72-h forecast. at each lead $\hat{y} = g_hat + \eta \cdot xgb$; the prediction overwrites the obs & lag1 channels of the next frame (autoregression). Intervals rescaled to $\mu g/m^3$.

Success criterion. engine RMSE < raw-model RMSE (else exit code 2 warning).

7.2 V2 — hybrid physics-bias correction

Chronological split. train = $[0.7 \cdot T]$, test = the remainder; features for $t \in [6, T)$ with 6 physics lags; label = $y_obs[t]$. No shuffling, no future data.

Model. HistGradientBoostingRegressor(max_iter = 200, learning_rate = 0.05, max_depth = 4); fits when ≥ 5 samples; physics is the leading feature (bias correction, not replacement).

Uncertainty. residuals calibrate a validation split; half-width = $1.96 \cdot \text{std}(\text{residuals})$ used for 72-h autoregressive intervals.

Ablation A–F isolates the marginal value of history / weather / fire / inversion blocks (Table in §5.10), scored by MAE/RMSE/ R^2 /MAPE/correlation/bias on the held-out split.

7.3 Shared metrics define ‘better’

$$MAE = \frac{1}{n} \sum |y - \hat{y}|, \quad RMSE = \sqrt{\frac{1}{n} \sum (y - \hat{y})^2}, \quad R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \quad \text{corr} = \rho(y, \hat{y}), \quad \text{bias} = \overline{\hat{y} - y}$$

8. V1 ↔ V2 comparison

Aspect	V1 — aqf_delhi	V2 — wrf_chem_delhi
Role	Production-shaped data platform + emulator + ML	Self-contained coupled forecast engine (SIH demo)
Data flow	Sources → validate → parquet → emulator → ML → API → dashboard	Closed loop: weather→...→feedback→next hour
Meteorology	Real GFS 0.25° GRIB2 (failover mirrors) or synthetic	Synthetic_weather (deterministic); live adapter exists
Fires	Live FIRMS (VIIRS 375 m) via API; FRP assigned to grid	Synthetic 33-fire stubble clusters (seed 11); live mode via FIRMS connector
Plume rise	Freitas-style 1-D integral, 12 layers	Same integral + Gaussian horizontal dispersion
Chemistry	O3 photochemical proxy, PM10 = 1.6·PM2.5, decay/wet loss	Fuller: O3 photolysis+titration, VOC→SOA, NOx→nitrate, hygroscopic growth
Coupling	One-way (weather→chemistry)	Two-way: aerosol feedback → PBL → transport → next hour
Scenarios	n/a (real/synthetic atmosphere)	3 tunable scenarios with peak PM bands
ML model	ST-GNN (T-GCN) + XGBoost residual + conformal 90%	HistGradientBoosting bias correction + ablation A-F
ML purpose	Spatio-temporal bias correction of the gridded model	Station hybrid correction; quantifies feature value
Training data	SyntheticDelhiExperiment (45/90 days) or live store	Physics forecast + synthetic observed profile
API	14 routes over artifacts (/observations, /fires, /gfs, /forecasts)	16 routes, richer payload + envelope
Dashboard	Unified home at / + /dashboard SPA (MapLibre)	Dashboard_v2 at / (scenario switch, AQI cards)
Version / tests	aqf_delhi 0.1.0 · API 0.4.0 · 47 tests	wrf_chem_delhi 2.0.0 · 22 V2 tests (69 total)
Mode label	data_source per partition	data_source + model_version in every envelope

What both share

One 21×23 / 4 km Delhi NCR domain; the same CPCB AQI breakpoints and bands; the same FRP→fuel(0.368)→EF emission mathematics; the same Freitas plume integral; the same PBL/ventilation/inversion physics core; honest labelling of synthetic data; and a clear path to plug in real WRF-Chem output.

9. Workflow summary (step-by-step)

Running V1

```
pip install -e .[dev,api,ml]
python scripts/run_ingest.py cpcb|firms|gfs # Module 1
python scripts/run_coupled.py [--init 2026-10-01T00Z] # Module 2 (demo or live GFS)
python scripts/run_module3.py demo # Module 3 (train + forecast)
uvicorn aqf_delhi.api.app:app --port 8000 # Module 4
open http://127.0.0.1:8000 # Module 5
```

Running V2

```
python -m wrf_chem_delhi.demo [--scenario {normal_winter,strong_inversion,stubble_plume}]
[--hours 72] [--fast]
python -m wrf_chem_delhi.server --port 8001 # API + dashboard
python -m pytest tests/ tests_v2/ # 69 tests
```

Flowchart reading guide — every arrow is a new artifact or state the next stage consumes; every state carries source/version labels so the user always knows whether numbers are live, cached, or demo.

10. Limitations & scientific honesty

Both systems are reduced-order emulators inspired by WRF-Chem — not operational WRF-Chem. No claim is made that the synthetic fields are a real forecast.

All demo numbers are deterministic synthetic data with `data\_source` labels (LIVE / CACHE / DEMO); real CPCB/FIRMS/GFS plug in behind the same interfaces.

Parameters such as DILUTION_FACTOR (0.25), fire_factor, feedback coefficients are tuning constants chosen to reproduce plausible Delhi NCR winter levels, not calibrated to observations.

V1 conformal intervals cover residual variability of the *synthetic* experiment; real-world coverage requires training on live data once the keys/feeds are provisioned.

The V2 adapter (`adapter/wrfchem.py`) shows the real-WRF drop-in contract (wrfout_d01*, wrfchemi_* markers) but the real-field mapping is a documented stub.

Acronyms referenced

WRF — Weather Research and Forecasting · FRP — Fire Radiative Power · VIIRS — Visible Infrared Imaging Radiometer Suite · SNPP — Suomi National Polar-orbiting Partnership · MODIS — Moderate Resolution Imaging Spectroradiometer · GRIB — GRIdded Binary · NCR — National Capital Region · PBL — Planetary Boundary Layer · AOD — Aerosol Optical Depth · SOA — Secondary Organic Aerosol · ST-GNN — Spatial-Temporal Graph Neural Network · T-GCN — Temporal Graph Convolutional Network · WRMSE — Weighted Root Mean Square Error · SPE — Spatial Pattern Error · MAD — Median Absolute Deviation · GRAP — Graded Response Action Plan · CAQM — Commission for Air Quality Management · MoEF — Ministry of Environment, Forest and Climate Change · OGD — Open Government Data · CAAQM — Continuous Ambient Air Quality Monitoring · NOMADS — NOAA Operational Model Archive and Distribution System · EPSG:4326 — WGS84 lon/lat CRS · SIH — Smart India Hackathon.

End of document. Generated directly from the repository sources (src/aqf_delhi, src/wrf_chem_delhi, configs/, tests/).